

resolution ΔE to necessarily identify individual eigenstates (which would be achieved if $\Delta E \rightarrow 0$). Crucially, such a notion is fully insensitive to the question of how many collections of the energy levels of the system populate the energy band ΔE , and in this way ensures a complete logical separation of energy-band thermalization from the energy levels.

Given that Eq. (8) holds, we will also show that in any complete basis of physical states $|\psi_k\rangle$, $\hat{A}$ thermalizes in almost all basis states for any time average larger than $2\pi/\Delta E$ (formally, Theorems 4.2 and 4.3 in Sec. 4):

$$\frac{1}{2T} \int_{-T}^T dt \langle \psi_k(t) | \hat{A} | \psi_k(t) \rangle \approx A_{\text{th}}, \quad \text{for any } T \gg \frac{2\pi}{\Delta E}. \quad (9)$$

Here, $x \gg y$ should be rigorously understood as $x > cy$ for some large constant $c > 1$. By “almost all” basis states, we mean that the fraction of basis states to which the above statement applies approaches 1 as our experimental resolution improves (i.e., $\epsilon \rightarrow 0$, $T\Delta E \rightarrow \infty$). We describe this result as applying to “almost all” basis states as this is the statement we expect to be accessible in experiments for large systems; however, with sufficient resolution, our results actually allow us to constrain thermalization in every single state in the Hilbert space as well (which we expect may be possible for sufficiently small systems). Eq. (9) applies independently of whether the energy levels are degenerate or show any kind of statistical correlations. Most significantly, limits such as $T \rightarrow \infty$ or $\Delta E \rightarrow 0$ are not *required* for this result, but may be taken if desired and allowed by the specific context. Our main physical emphasis here is that knowing the properties of observables even at a low resolution corresponding to an energy band ΔE is sufficient to establish thermalization over finite or infinite time averages in arbitrary physical states, without requiring any higher resolution knowledge of the energy levels such as in Eq. (2).

In addition to time averages, one is also interested in showing that the observable attains thermal equilibrium at almost all times in a finite time range $t \in [-T, T]$. Unlike the distinction between eigenstate thermalization in Eq. (2) and the full statement of ETH in Eq. (3), where ETH assumes a separate off-diagonal structure for thermal equilibrium, we will show that the same notion of energy-band thermalization is also sufficient to describe thermal equilibrium. Specifically, we will require (in Sec. 7) that a cloned version of $\hat{A}$ in a doubled Hilbert space (each with dynamics generated by $\hat{H}$) also satisfies energy-band thermalization for the doubled Hamiltonian:

$$[\hat{A} \otimes \hat{A}]_{\Delta E} \approx A_{\text{th}}^2 \hat{1} \otimes \hat{1}. \quad (10)$$

We note that this cloning of the operator is a mere formal device, and such properties can be experimentally established entirely within a single copy of the system. Eq. (10) is sufficient for the observable to attain thermal equilibrium in the following sense:

$$\langle \psi_k(t) | \hat{A} | \psi_k(t) \rangle \approx A_{\text{th}} \text{ for almost all } t \in [-T, T], \text{ for any } T \gg 2\pi/\Delta E. \quad (11)$$

Here, “almost all t ” should be interpreted as “all t except a set whose length is smaller than T/c_2 for some large constant $c_2 > 1$ ”.

This “cloning” strategy may be regarded as a quantum version of a theorem in classical ergodic theory [6], which states that a dynamical system is weakly mixing (which loosely corresponds to thermal equilibrium) if and only if its “cloned” version with two copies of the system is ergodic (which loosely corresponds to time-averaged thermalization). This is especially convenient for us, as it allows us to use the same conceptual notion of energy-band thermalization (but applied to the observable vs. its cloned version) to address both time-averaged thermalization and instantaneous thermal equilibrium over finite timescales.

Collectively, Eqs. (8), (9), (10) and (11) establish a fully quantum connection between properties of observables with a finite energy resolution and their thermalization over finite time scales.

1.2.2 Bypassing energy levels with finite-time autocorrelators

We will now describe how energy-band thermalization [Eqs. (8), (10)] is advantageous compared to conventional statements of ETH [Eqs. (2), (3)] not only in the ability to access finite time scales,